COURSE NAME

SOFTWARE ENGINEERING

CSC 3114

(UNDERGRADUATE)

CHAPTER 4

AGILE SOFTWARE DEVELOPMENT

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WHAT IS AN AGILE METHOD?

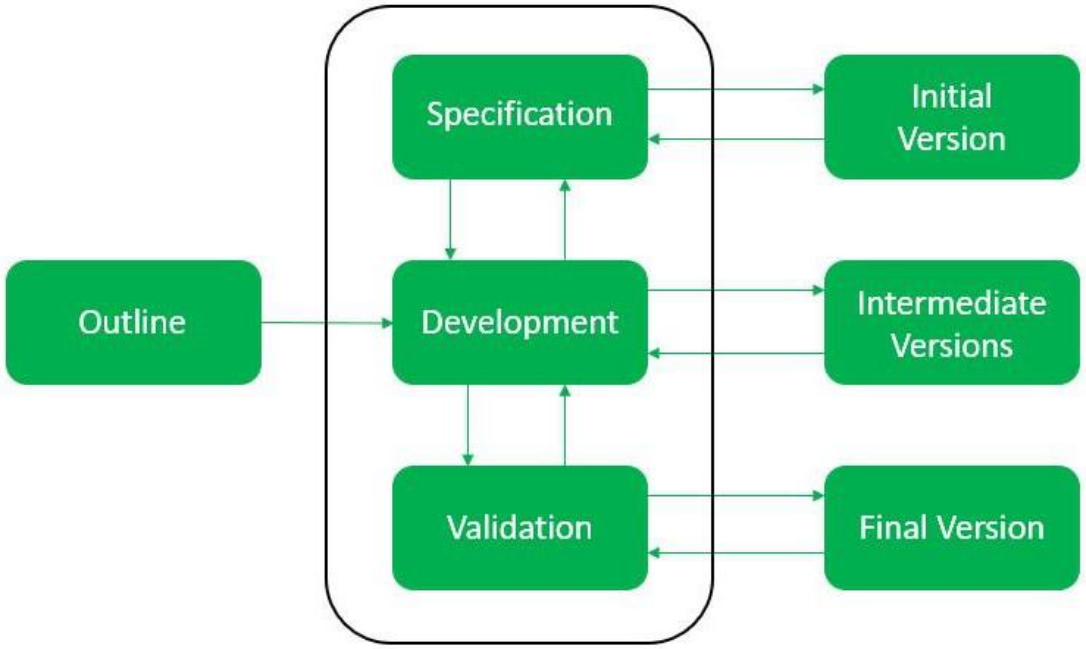
- Agile methods are considered
  - Lightweight (do not concentrate on the whole software development at once)
  - People-based rather than Plan-based
- Several agile methods
  - No single agile method
  - Different agile methods can be combined in software development (Hybrid)

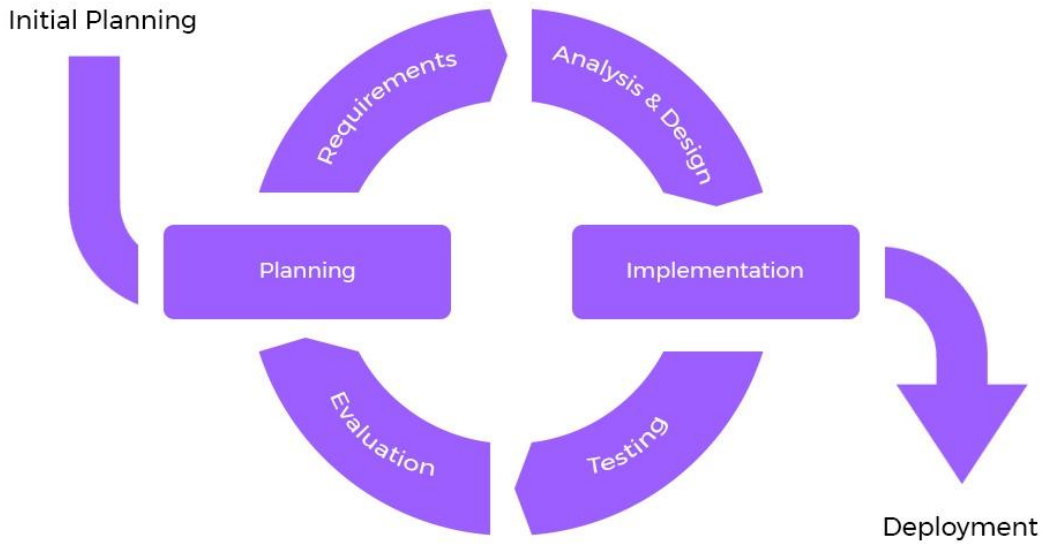
AGILE MODEL

- Subset of iterative and evolutionary methods

Iterative Products

- Each iteration is a self-contained, mini-project with activities that span requirements analysis, design, implementation, and test
- Leads to an iteration release (which may be only an internal release) that integrates all software across the team and is a growing and evolving subset of the final system
- The purpose of having short iterations is so that feedback from iterations N and earlier, and any other new information, can lead to refinement and requirements adaptation for iteration N + 1





EVOLUTIONARY + ITERATIVE MODEL = AGILE

AGILE METHODS VS. PAST ITERATIVE METHODS

- A key difference between agile methods and past iterative methods is the length of each iteration.

- In the past, iterations might have been three or six months long.
- In agile methods, iteration lengths vary between one to four weeks and intentionally do not exceed 30 days.
- Research has shown that shorter iterations have lower complexity and risk, better feedback, and higher productivity and success rates.

TIMEBOX & SCOPE

- The pre-determined iteration length serves as a timebox for the team.
- Scope (set of tasks) is chosen for each iteration to fill the iteration length.
- Rather than increase the iteration length to fit the chosen scope, the scope is reduced to fit the iteration length.

AGILE VS. PLAN DRIVEN PROCESS

Agile Process	Plan Driven Process
Small products and teams; scalability limited	Large products and teams; hard to scale down
Inappropriate for safety-critical products because of frequent changes	Handles highly critical products
Good for dynamic, but expensive for stable environments.	Good for stable, but expensive for dynamic environments
Require experienced Agile personnel throughout	Require experienced personnel only at start if stable environment
Personnel succeed on freedom and chaos	Personnel succeed on structure and order

HUMAN FACTORS IN AGILE DEVELOPMENT

- Skill / Capability
- Common focus
- Collaboration
- Decision-making ability
- Fuzzy problem-solving ability
- Mutual trust and respect
- Self-organization

AGILE METHODS

- Extreme Programming (XP)
- Scrum
- Dynamic Systems Development Method (DSDM)
- Feature-Driven Development (FDD)
- Crystal Methods
- Lean Development (LD)
- Adaptive Software Development (ASD)

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- Kelly, J. C., Sherif, J. S., & Hops, J. (1992). An analysis of defect densities found during software inspections. *Journal of Systems and Software*, *17*(2), 111-117.
- Bhandari, I., Halliday, M. J., Chaar, J., Chillarege, R., Jones, K., Atkinson, J. S., & Yonezawa, M. (1994). In-process improvement through defect data interpretation. *IBM Systems Journal*, *33*(1), 182-214.